Explainable AI · XAI

Explainable Machine Learning

Shapley Values and SHAP from a Finance Professional's Perspective

Explainable Machine Learning — Shapley Value · SHAP · XGBoost

A score alone is not enough. How to explain, validate, and improve why a model made its decision

Restructured from the original lecture (2019), updated with research and regulation as of September 2026

Three Messages of This Lecture

1

We need to know the reason behind the score

A fraud probability of 0.5 or a default probability of 55% alone lets us neither explain to customers, validate the model, nor improve it

2

The Shapley value is the unique fair way to allocate contributions

Only one allocation satisfies the four axioms of cooperative games, and SHAP carries it over to feature contributions

3

XGBoost's built-in importance changes rank with each option

weight, cover, and gain are inconsistent. SHAP covers global, individual, interaction, and monitoring views in one framework

Explainability = a common language for trust · validation · regulatory compliance

Lecture Outline — Seven Parts

Motivation → limits of existing importance → theory (Shapley) → method (SHAP) → lab → advanced → practice

Part	Topic	Key question
1	Why explainability?	A model that predicts well may do so for the wrong reasons — wolves, fraud, loan rejections
2	Limits of existing feature importance	Why do XGBoost's weight, cover, and gain give different rankings?
3	Shapley values	How do we define a fair allocation of contributions mathematically?
4	SHAP	Bringing Shapley values to model explanation, and computing them fast
5	XGBoost + SHAP lab	Reading a model with summary and dependence plots (Adult income · synthetic credit data)
6	Advanced SHAP	Interaction values and model monitoring (mortality · surgery duration models)
7	Financial practice and limits	Rejection reasons · regulation · SHAP pitfalls · recent trends

01

PART 1 · Motivation

Why explainability?

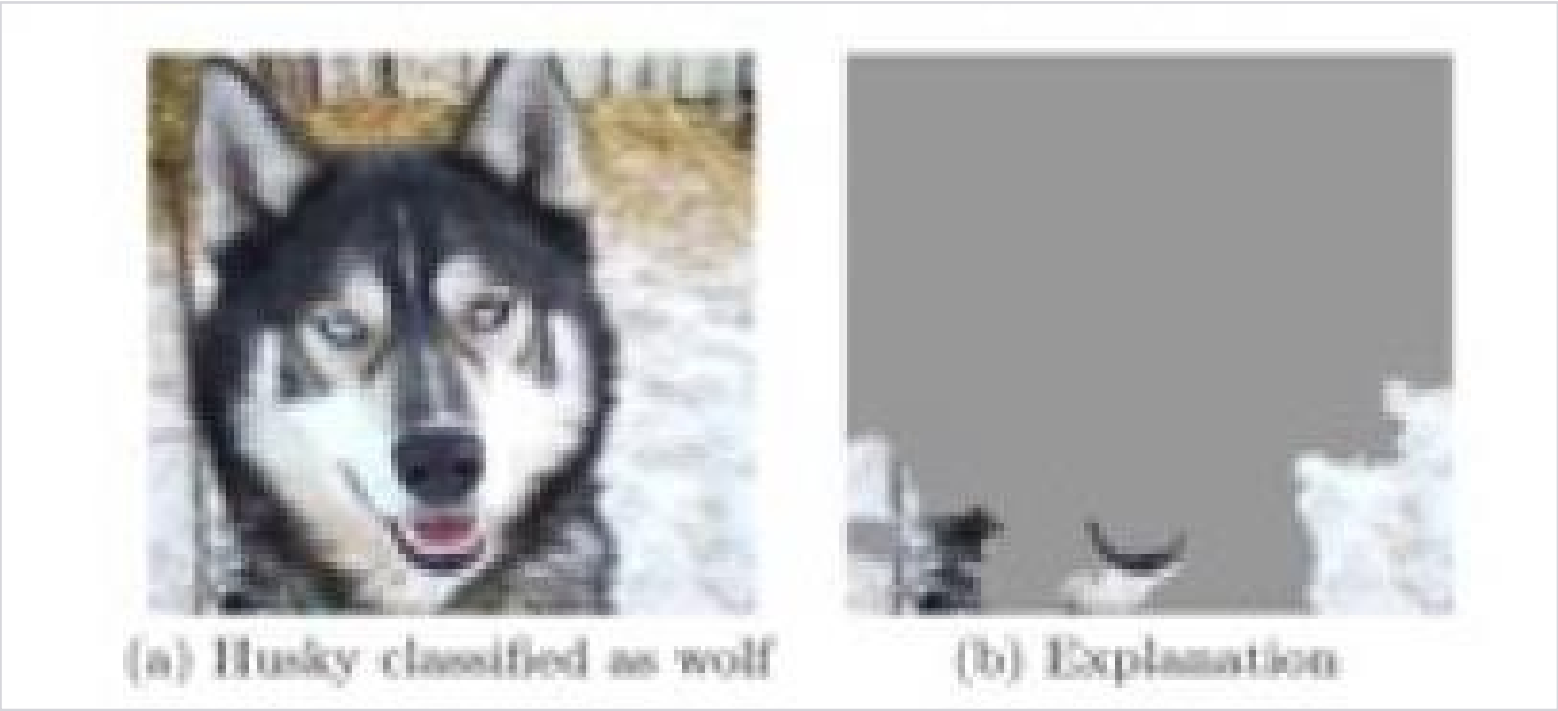
A model that predicts well may do so for the wrong reasons

Motivation for Explainability ① Wolf vs. Husky

Machine learning classified them remarkably well, but the reason lay somewhere unexpected

Episode · LIME (2016)

- The researchers deliberately chose 20 training images so that only wolf photos had snow backgrounds
- Opening the explanation revealed that the model judged "wolf" from the white snow, not the animal
- Graduate students who trusted the model after seeing the explanation: 10 of 27 → 3



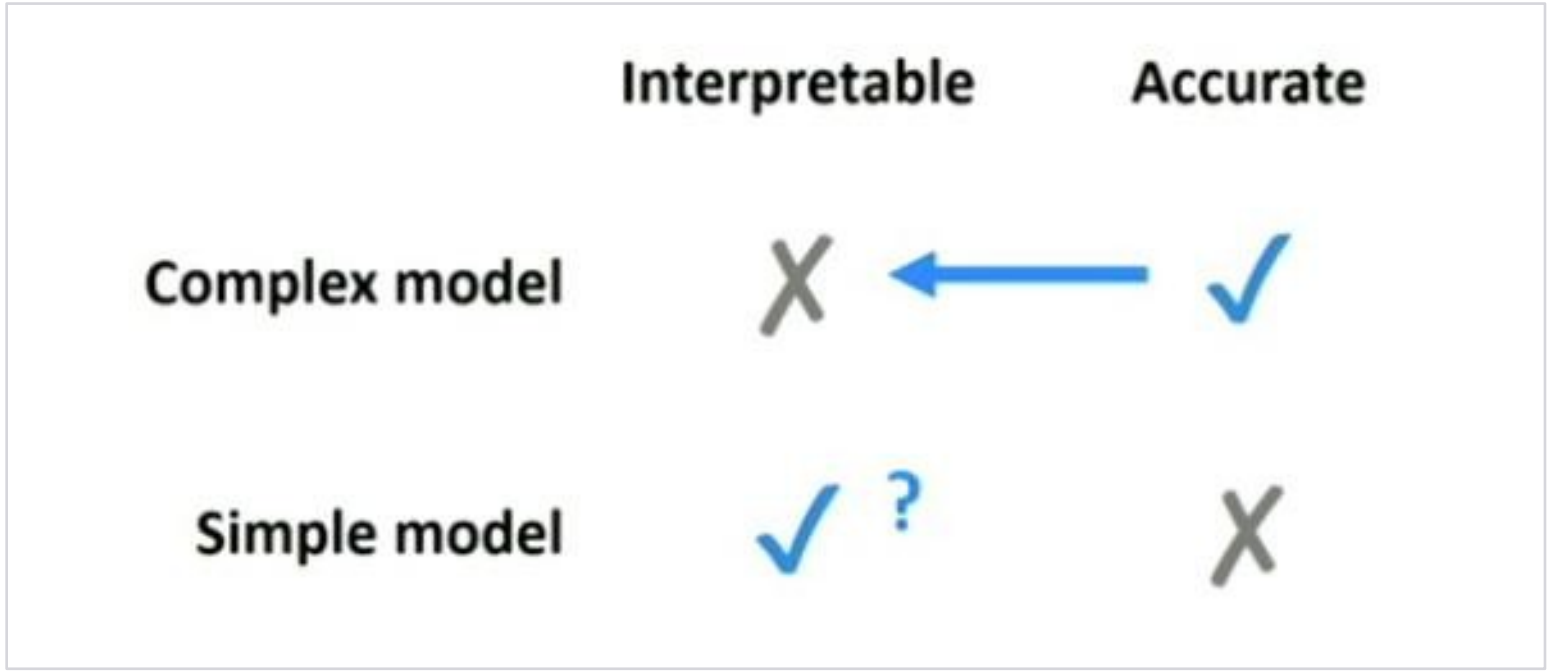
Ribeiro·Singh·Guestrin, “Why Should I Trust You?”, KDD 2016

Motivation for Explainability ② Simplicity vs. Interpretability

Complex models predict well, but as black boxes they are hard to interpret

How to read

- Arrow: the goal of XAI is to restore interpretability to accurate complex models
- The "?" on simple models: a logistic regression with hundreds of variables is practically hard to interpret too
- There is a counterargument: use interpretable models from the start for high-stakes decisions (Rudin, 2019) → Part 7



Motivation for Explainability ③ Credit Card Fraud Detection

What is the probability of credit card fraud? Where on earth does this probability come from?

Questions from the field

- Approval or rejection is decided within seconds at the moment of payment
- The model outputs only a single probability
- Investigators and call centers must answer "Why was it blocked?"

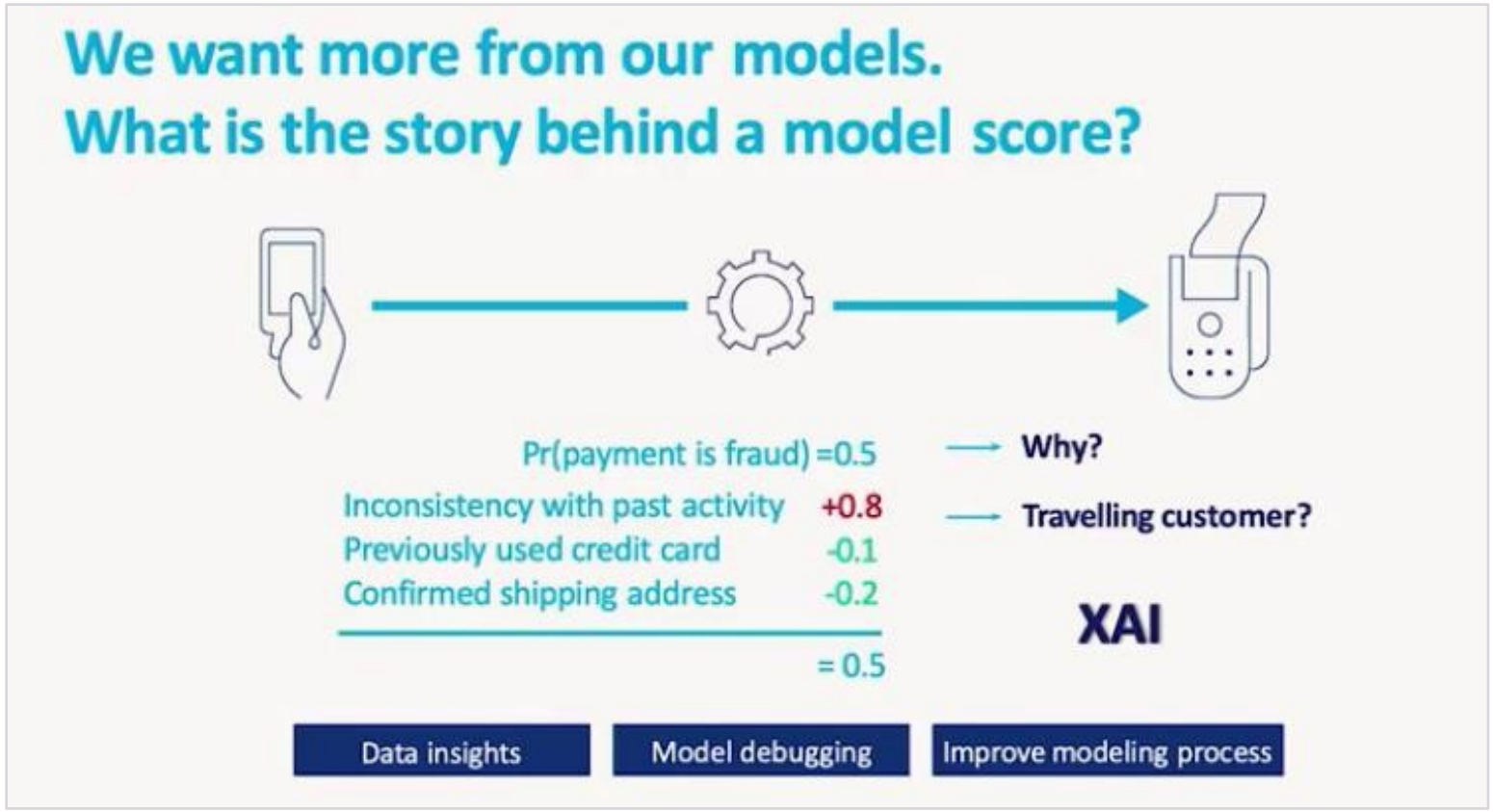


Motivation for Explainability ④ The Story Behind the Score

Knowing only the score is not enough. We must know the reasons and context to explain, validate, and improve

What XAI provides

- Decomposition: inconsistency with past activity +0.8, previously used card −0.1, confirmed shipping address −0.2 → total 0.5
- If the largest factor is "behavior unlike the past," the customer may be travelling → a rule to reduce false positives
- Three benefits: data insights · model debugging · improving the modeling process

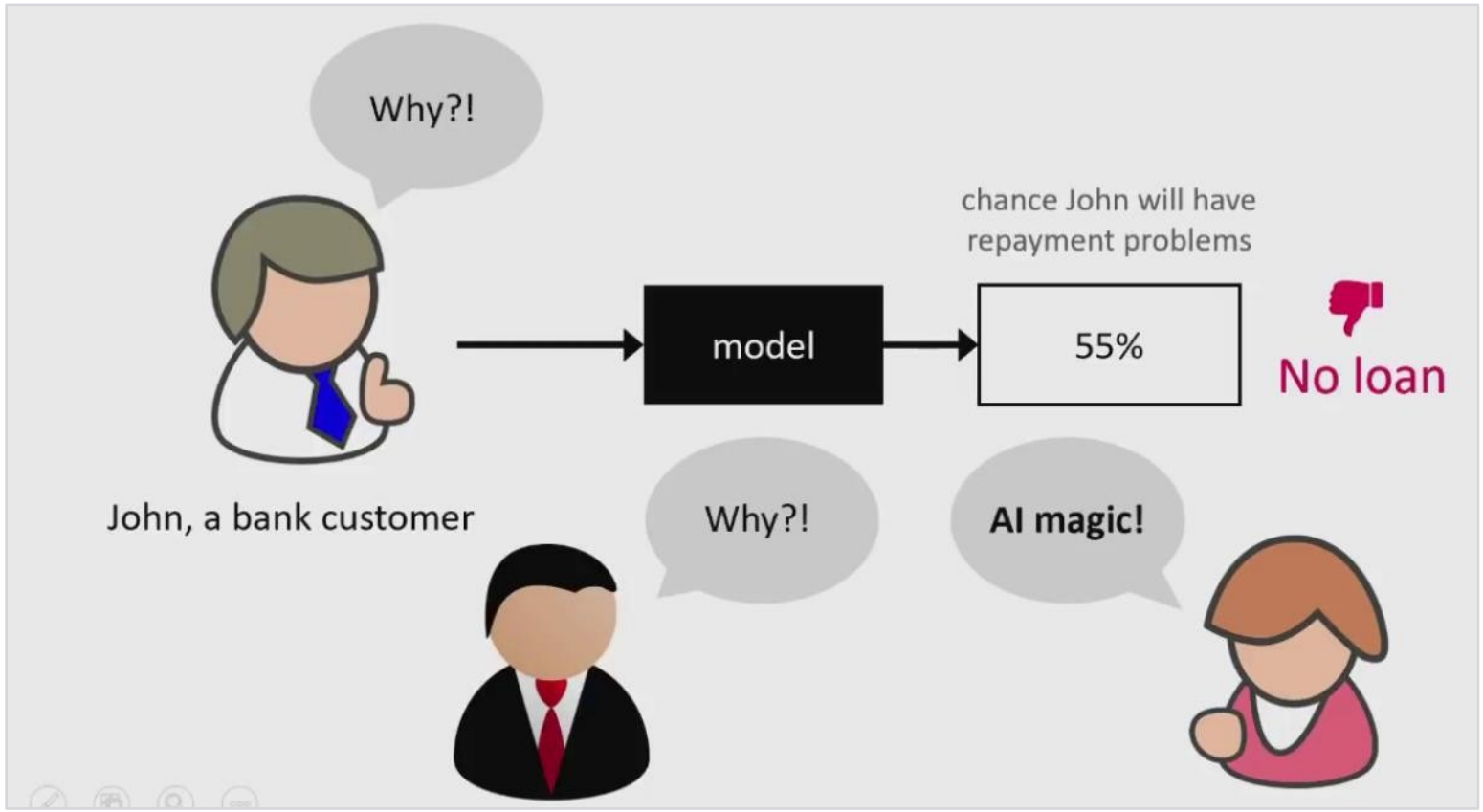


Motivation for Explainability ⑤ How to Explain a Loan Rejection

The EU GDPR requires explaining the reasons for automated decisions

The issue

- "The AI said so" cannot be the answer to a customer asking "Why was I rejected?"
- GDPR Art. 22 · Art. 15(1)(h): "meaningful information about the logic involved" in automated decisions
- The reason for rejection may be a discriminatory factor such as race or gender — impossible to check without explanation



Definitions and Taxonomy of Interpretability

Terms are used loosely across the literature; this lecture follows the distinctions below

Distinction	Meaning	Example
Interpretability	The degree to which a person can understand the model structure itself	Linear regression coefficients, shallow trees
Explainability	Rendering a complex model's decisions into an understandable form after the fact	SHAP, LIME
Global vs. local	Model-wide tendencies vs. the reason for a single prediction	Feature importance vs. one customer's rejection reasons
Intrinsic vs. post-hoc	The model explains itself vs. a separate tool explains it after training	GAM·EBM vs SHAP·PDP
Model-specific vs. model-agnostic	Uses a specific structure vs. uses only inputs and outputs	TreeSHAP vs KernelSHAP·LIME

A Map of XAI Methods

SHAP is one method among many — placed by scope (global · local) and model dependence

Method	Scope	Model	Key idea
Permutation importance	Global	Agnostic	How much does performance drop when a feature is shuffled?
PDP · ICE	Global · individual	Agnostic	Track average and individual prediction changes as a feature value varies
ALE	Global	Agnostic	A PDP alternative with less bias under correlated features
LIME	Local	Agnostic	Approximate the neighborhood of a prediction with a simple linear model
SHAP	Local → global	Agnostic · specific	Allocate contributions by Shapley values; they sum to the prediction
Counterfactual explanation	Local	Agnostic	What would have had to differ for the outcome to change?
EBM · GAM	Global	Intrinsic	Trained with an additive structure from the start and read directly

Case · Apple Card Credit Limit Controversy (2019–2021)

Even with a finding of no discrimination, distrust of the "black box" remained

20×
Claimed limit gap (tweet, Nov 2019)

About 400,000
New York applicants reviewed by the regulator

No violation
NYDFS conclusion (Mar 2021)

Timeline

- A founder tweeted "my wife's limit is 1/20 of mine"; an Apple co-founder raised a similar case
- The New York State Department of Financial Services (NYDFS) investigated the issuer, Goldman Sachs
- Men and women with the same credit characteristics got similar outcomes — no evidence of discrimination

Lessons

- The report also noted customers' frustration with a "black-box algorithm"
- Without a system to explain individual decisions in customer language, reputational costs arise
- The issuer later strengthened guidance on the review process and appeal procedures

NYDFS, Report on Apple Card Investigation (2021.3)

02

PART 2 · Existing Feature Importance

Limits of XGBoost's Built-in Importance

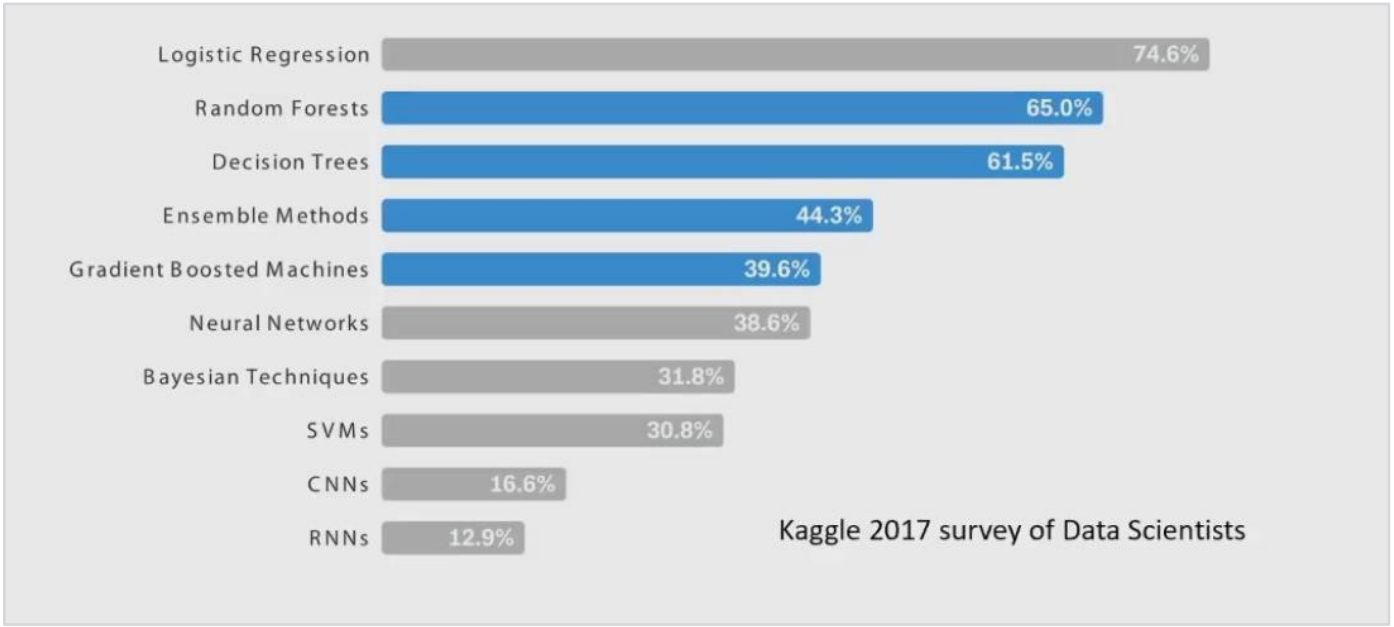
Change the option and the ranking changes — what should we trust?

Tree Models Are the Most Popular Complex Models

The complex models most used in practice are tree-based — so explaining tree models matters

How to read · Supplement

- Blue bars: tree-based models (random forest · decision tree · ensemble · GBM)
- The original slide's figures differ from the official 2017 Kaggle report (LR 63.5% · DT 49.9% · RF 46.3%), so they appear to reflect a specific respondent group
- On tabular data, tree boosting still beats deep learning (Grinsztajn et al., NeurIPS 2022)



Kaggle, State of Data Science & ML Survey 2017

XGBoost Feature Importance — Three Options

xgboost.plot_importance(model, importance_type=...)

Option	Definition	Bias to watch
weight	Number of times the feature is used to split data across all trees	Favors continuous features with many values
cover	Average number of training samples (sum of Hessians) passing through the feature's splits	Gives weight only to splits that divide many samples
gain	Average reduction in training loss from splits on the feature	Tends to overrate splits lower in the tree

Practice note · With total_gain · total_cover (totals) there are five. plot_importance defaults to weight, while the scikit-learn API's feature_importances_ defaults to gain — the "important variables" in a report depend on which one you use

What Do gain and cover Measure?

XGBoost picks splits using the first and second derivatives of the loss (G, H) — and importance comes from there too

$$\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma$$

$$\text{Cover} = \sum_{i \in \text{node}} h_i, \qquad h_i = \frac{\partial^2 \ell(y_i, \hat{y}_i)}{\partial \hat{y}_i^2}$$

Notation

- G, H: sums of the first derivatives (gradient) and second derivatives (hessian) of samples in a node
- λ, γ: regularization terms on leaf weights and the number of leaves

What cover means

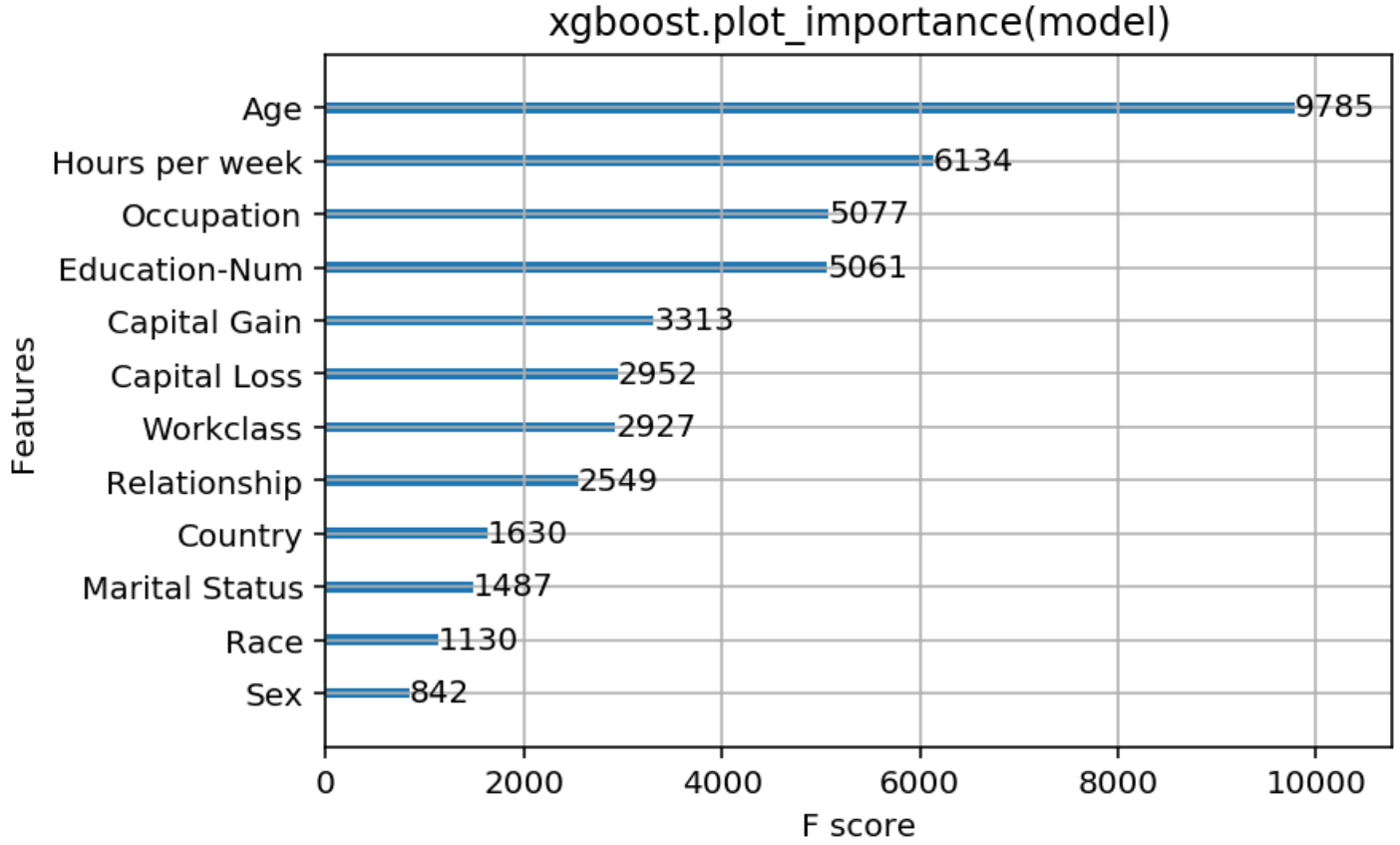
- For squared error, h = 1 → cover = number of samples
- For logistic loss, h = p(1−p) → confident samples count less

Global Feature Importance Measures — XGBoost Defaults

UCI Adult census data: an XGBoost model predicting whether annual income exceeds $50K

What to look at

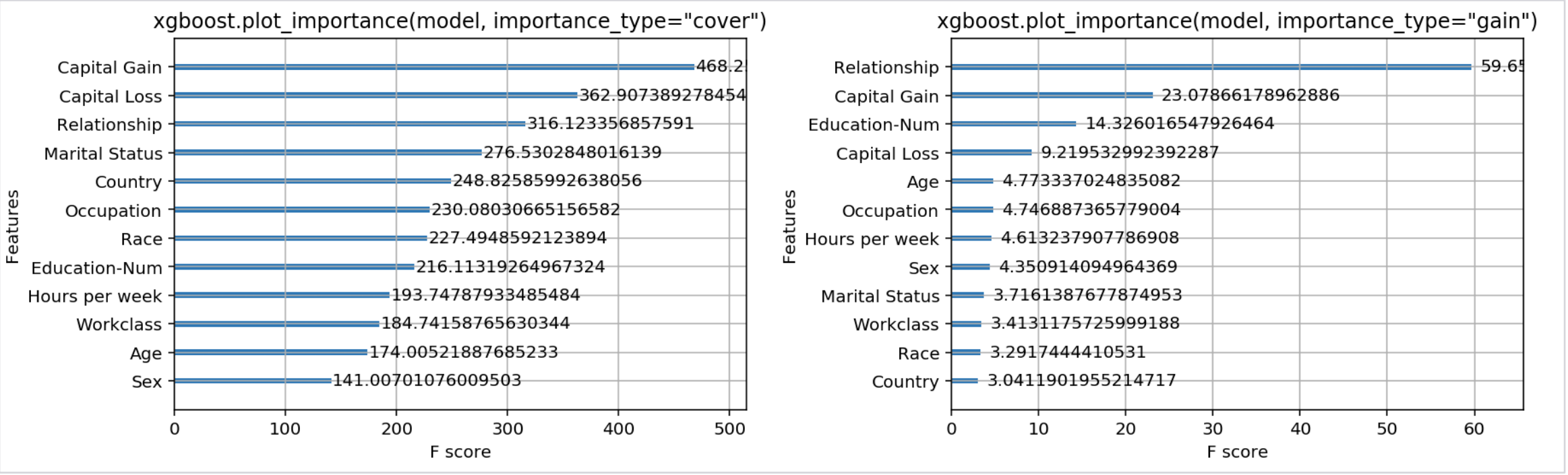
- plot_importance defaults to weight (number of splits)
- Age is by far #1, Sex is last
- Is Age really the most important? Let's change the option



Lundberg, "Interpretable Machine Learning with XGBoost" (2018)

The Importance Ranking Changes with the Option

#1 is Age under weight, Capital Gain under cover, Relationship under gain — same model, three different answers

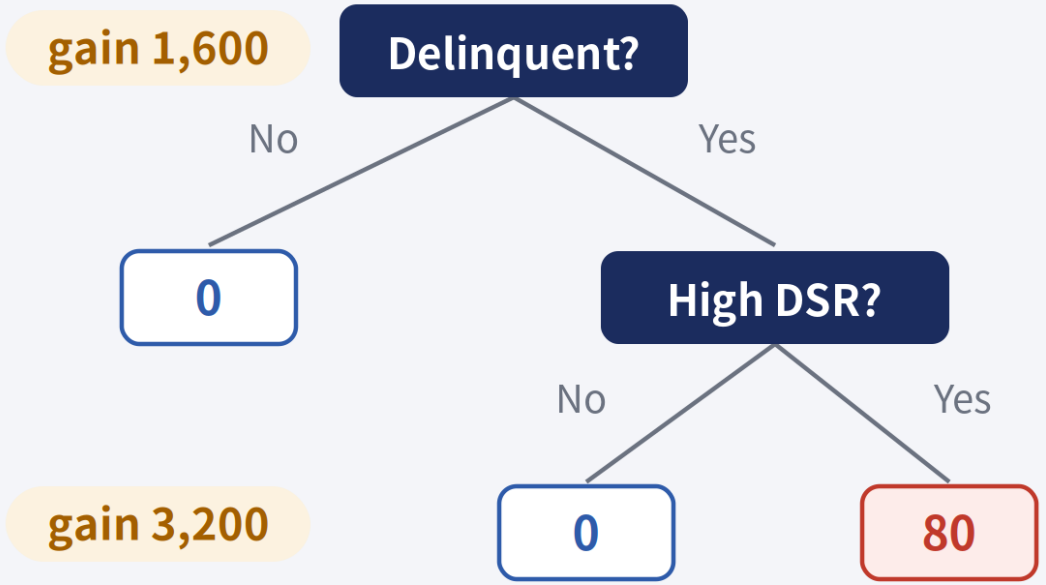


▶ Disappointingly, the importance ranking depends on the option — which one should we report?

Why It Wobbles — Just Changing the Split Order Flips gain

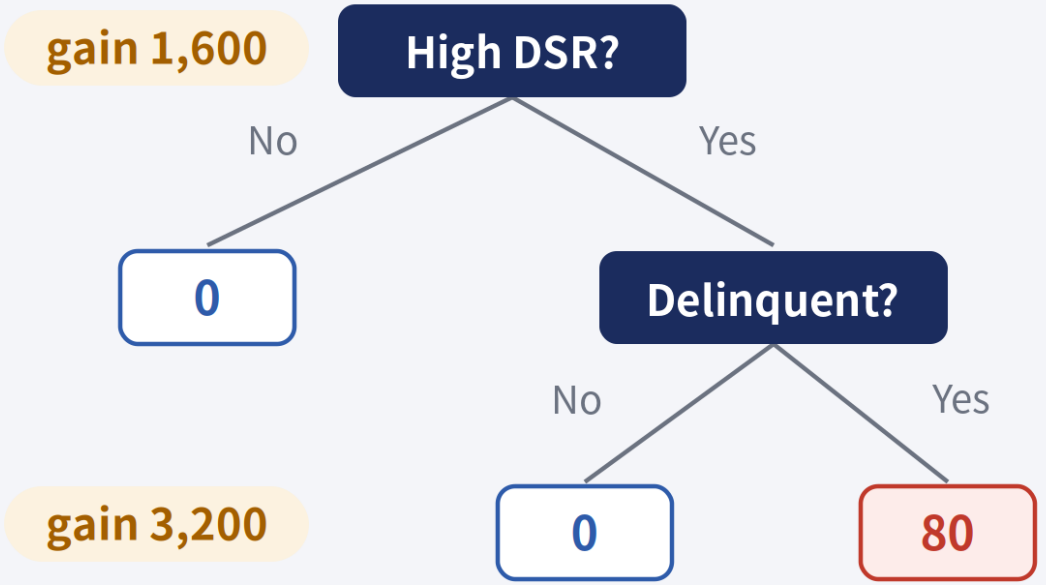
Both are the same function, "80 points only if delinquent AND high DSR" — gain follows the order, SHAP stays the same

Tree ① Split on delinquency first



gain: delinquency 1,600 · DSR 3,200
SHAP: delinquency 30 · DSR 30

Tree ② Split on high DSR first



gain: delinquency 3,200 · DSR 1,600
SHAP: delinquency 30 · DSR 30

Two Conditions for Good Importance — Consistency and Accuracy

Lundberg (2018): weight, cover, and gain violate consistency; only Shapley values (TreeSHAP) satisfy both conditions

$$f'_x(S) - f'_x(S \setminus i) \;\geq\; f_x(S) - f_x(S \setminus i) \;\; \forall S \quad \Longrightarrow \quad \phi_i(f', x) \geq \phi_i(f, x)$$

Consistency

- If a model changes to rely more on a feature, that feature's importance must not decrease

Local accuracy

- The feature contributions must sum exactly to the change in prediction relative to the base value

Customer (delinquent · high DSR)	Delinquency SHAP	High-DSR SHAP	Prediction – mean
Model A: 80 only under AND	30	30	80 − 20 = 60
Model B: A + 10 for high DSR alone	30	35 ↑	90 − 25 = 65

Reproduction with Synthetic Credit Data — Experimental Design

Compare four importance measures on data whose true model is known

Experimental design (synthetic data)

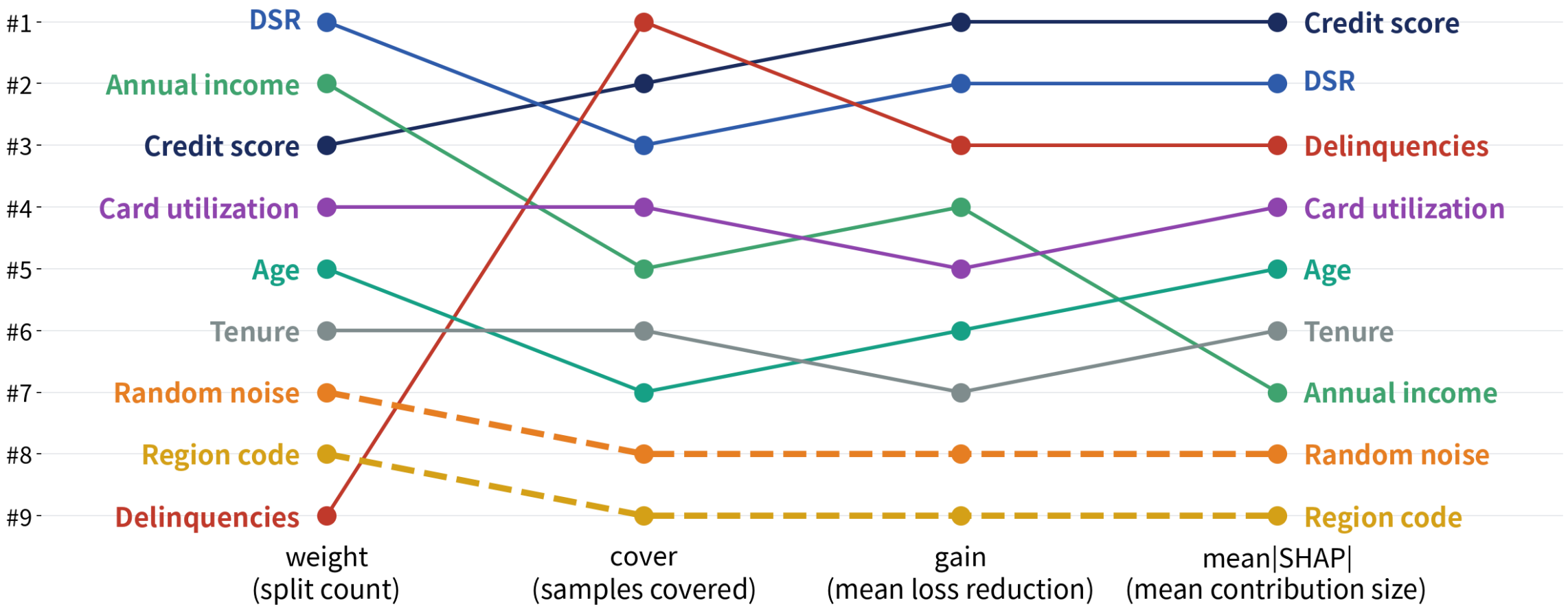
- 30,000 loans, defaults generated from a logistic true model (default rate 12.7%)
- Main true drivers: credit score · DSR · delinquencies · card utilization; DSR effect is 1.8× when annual income is below KRW 40 million
- Random noise · region code: unrelated to default (ground truth = zero effect)
- Train XGBoost (400 trees, depth 4), then compare weight · cover · gain · mean|SHAP|

Observations

- weight ranks delinquencies (values only 0 · 1 · 2 …) 9th — not because it is unimportant, but because it has few chances to split
- cover lifts the same delinquencies to 1st
- gain and SHAP have similar top ranks. But only SHAP guarantees consistency and decomposes each case

Same Model, Four Rankings

Delinquencies: 9th by weight → 1st by cover → 3rd by gain · SHAP (dashed = variables unrelated to default)



03

PART 3 · Game Theory

Shapley Value

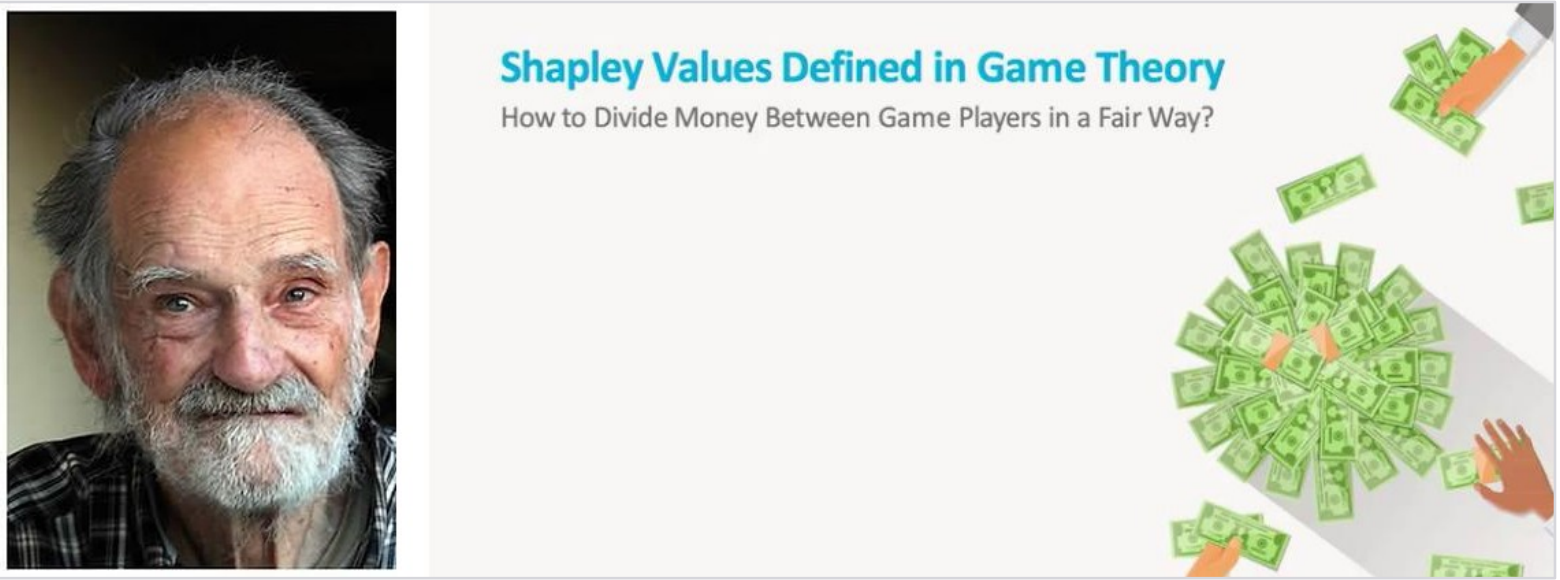
How to fairly divide each player's contribution in a cooperative game

Shapley Values — Fair Allocation from Game Theory

A value measuring each player's contribution in a cooperative game. In machine learning, the players are features

Background

- Lloyd Shapley (1923–2016), 1953 paper "A Value for n-Person Games"
- 2012 Nobel Prize in Economics (shared with Alvin Roth) — theory of stable allocations and market design
- Question: how do we fairly divide money earned together among everyone?



Cooperative Games ↔ ML Explanation — Matching the Terms

Reread the problem as dividing the gap between an individual prediction and the average prediction "fairly" among the features

Cooperative game	ML explanation
Player set N	Feature set — credit score, DSR, income …
Coalition $S \subseteq N$	Subset of features whose values are known
Value function v(S)	Expected prediction when only S is known
Total to divide $v(N) - v(\varnothing)$	This customer's prediction f(x) − average prediction E[f(X)]
Allocation φ(i)	Contribution of feature i = SHAP value

Shapley value = the average of what feature i adds (marginal contribution) over all joining orders

Definition of the Shapley Value

Take a weighted average, over all possible cases, of the value added when feature i joins coalition S

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \Big[v\big(S \cup \{i\}\big) - v(S) \Big]$$

Same value, in order-averaging form

$$\phi_i(v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} \Big[v\big(P_i^\pi \cup \{i\}\big) - v\big(P_i^\pi\big) \Big]$$

P: the set of features that joined before i in order π

How to read

- Brackets: the marginal contribution when i joins
- Weight: the number of orderings before and after S — every joining order is treated equally
- That is, the average of i's marginal contribution over n! orders

Four Axioms of Fairness

The Shapley value is the only allocation rule satisfying all four axioms (Shapley, 1953)

Axiom	Meaning	In credit scoring terms
Efficiency	Sum of contributions = total to divide	Sum of contributions = this customer's default probability − average default probability
Symmetry	Equal roles get equal shares	Two variables with identical effects get the same contribution
Dummy (null player)	No contribution means zero	A variable the model does not use contributes zero
Additivity	Share of the sum of two games = sum of each share	SHAP of 400 trees = sum of per-tree SHAP

$$\sum_{i=1}^{n} \phi_i = v(N) - v(\varnothing)$$

Example ① Dividing Customer A's 30% Default Probability

Customer A: delinquent · high DSR · high income (hypothetical figures)

Known features S	Predicted default probability v(S)
None (⌀)	10%
Delinquency	22%
DSR	16%
Income	8%
Delinquency + DSR	34%
Delinquency + income	19%
DSR + income	13%
All three features (N)	30%

How to read

- v(⌀) = 10%: when nothing is known = the average customer
- v(N) = 30%: the model's prediction for customer A
- Share to divide: 30 − 10 = 20%p
- Knowing income actually lowers it (22% → 19%) — contributions can be negative

Example ② Averaging over Six Joining Orders

For each order, record the increase in default probability (%p) as each feature joins, then average by column

Joining order	Delinquency	DSR	Income
Delinquency → DSR → income	+12	+12	−4
Delinquency → income → DSR	+12	+11	−3
DSR → delinquency → income	+18	+6	−4
DSR → income → delinquency	+17	+6	−3
Income → delinquency → DSR	+11	+11	−2
Income → DSR → delinquency	+17	+5	−2

Delinquency φ	DSR φ	Income φ	Total = 30 − 10
+14.5%p	**+8.5%p**	**−3.0%p**	**+20.0%p**

The Problem Is Computation

With M features there are 2^M coalitions — unusable in practice as is

$$\underbrace{O\left(T\,L\,2^{M}\right)}_{\text{brute force}} \longrightarrow \underbrace{O\left(T\,L\,D^{2}\right)}_{\text{TreeSHAP}}$$

Problem

- 30 features → about 1 billion coalitions (2^30)
- Each explanation would require hundreds of millions of model evaluations
- T: number of trees, L: max number of leaves, D: max depth

Solution

- Sampling approximations: KernelSHAP, permutation sampling
- Exploiting structure: TreeSHAP walks the trees to compute exact values in polynomial time
- For an XGBoost model, tens of thousands of cases take seconds

04

PART 4 · SHAP

SHAP — Explaining Models with Shapley Values

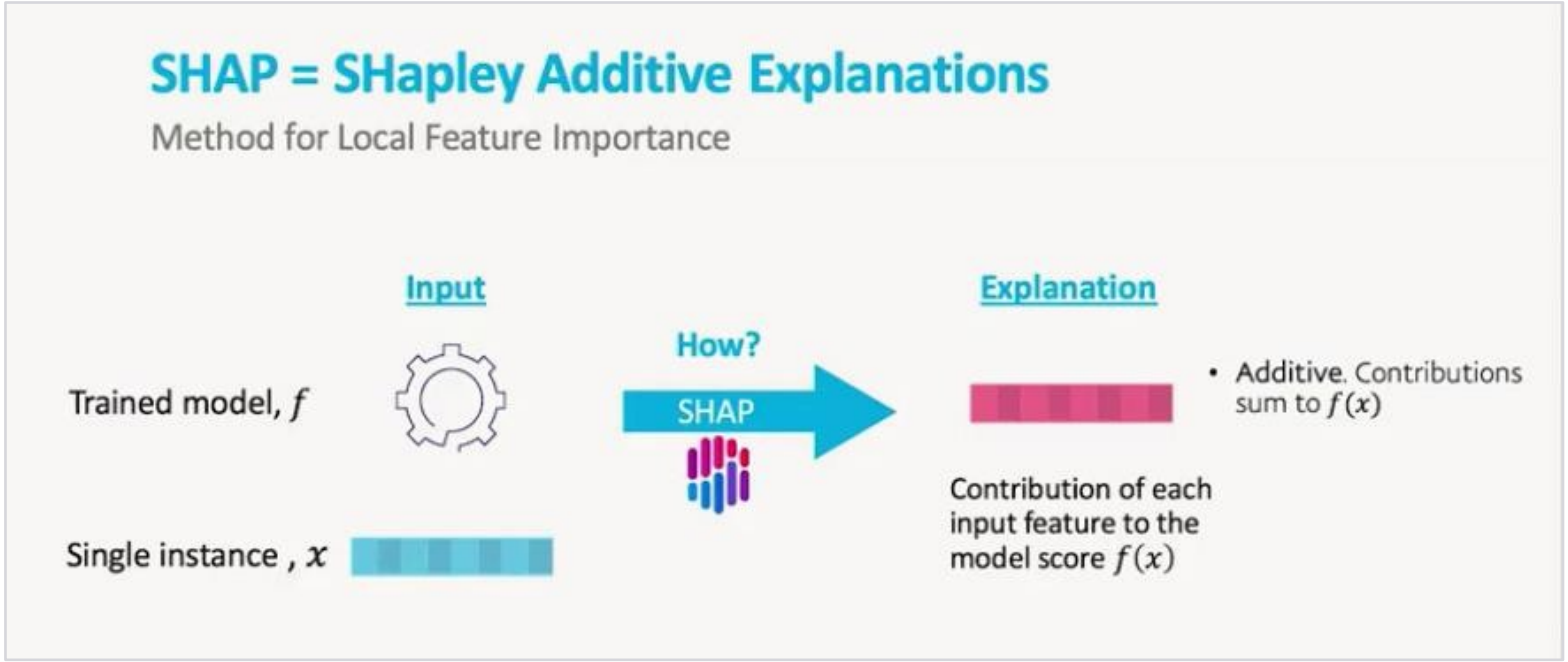
SHapley Additive exPlanations (Lundberg & Lee, NeurIPS 2017)

SHAP = SHapley Additive exPlanations

Shapley values are consistent and additive. SHAP builds on them to extract local feature importance

Structure

- Input: a trained model f and one observation x to explain
- Output: a vector of per-feature contributions
- Additivity: the base value plus all contributions equals f(x)



SHAP's Decomposition and Three Properties

Prediction = average prediction (base value) + sum of per-feature contributions

$$f(x) = \underbrace{\mathbb{E}\big[f(X)\big]}_{\phi_0 :\ \text{base value}} + \sum_{i=1}^{M} \phi_i(x)$$

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i \, z'_i, \qquad z' \in \{0,1\}^M$$

Property	Meaning	Basis
Local accuracy	Base value + sum of contributions = prediction	Same as the efficiency axiom
Missingness	A feature absent from the input contributes 0	Corresponds to the dummy axiom
Consistency	If the model relies more on a feature, its contribution does not decrease	The Shapley value is the only solution satisfying all three

How to Mimic a "Missing Feature" — Choosing the Value Function

There are two ways to define v(S), and with correlated features the answers differ

$$v_x(S) = \mathbb{E}\left[f(X) \mid X_S = x_S\right]$$

$$v_x(S) = \mathbb{E}_{X_{\bar{S}}}\left[f(x_S, X_{\bar{S}})\right]$$

Observational (conditional) — true to the data

- Fill in the remaining features from the distribution that co-occurs with x_S
- Preserves the correlation structure, but contribution leaks to features the model does not use
- tree_path_dependent, used when TreeExplainer gets no background data, is close to this

Interventional — true to the model

- Swap in the remaining features wholesale from background data (data=...)
- Only features the model actually uses receive contributions
- Unrealistic combinations such as age 22 with 30 years of tenure can arise

Chen·Janizek·Lundberg·Lee, “True to the Model or True to the Data?” (2020)

The SHAP Estimator Family

Choose the estimator to fit the model type — for tree models, TreeSHAP is the answer

Estimator	Target model	Approach	Speed
TreeSHAP	XGBoost · LightGBM · CatBoost · random forest	Exact computation via tree structure, $O(TLD^2)$	Fast
KernelSHAP	All models	Approximated by weighted linear regression with Shapley weights	Slow
Linear · Exact	Linear models · models with few features	Analytic computation · full enumeration	Fast · small scale
DeepSHAP · GradientSHAP	Neural networks	DeepLIFT · gradient-based approximation	Medium

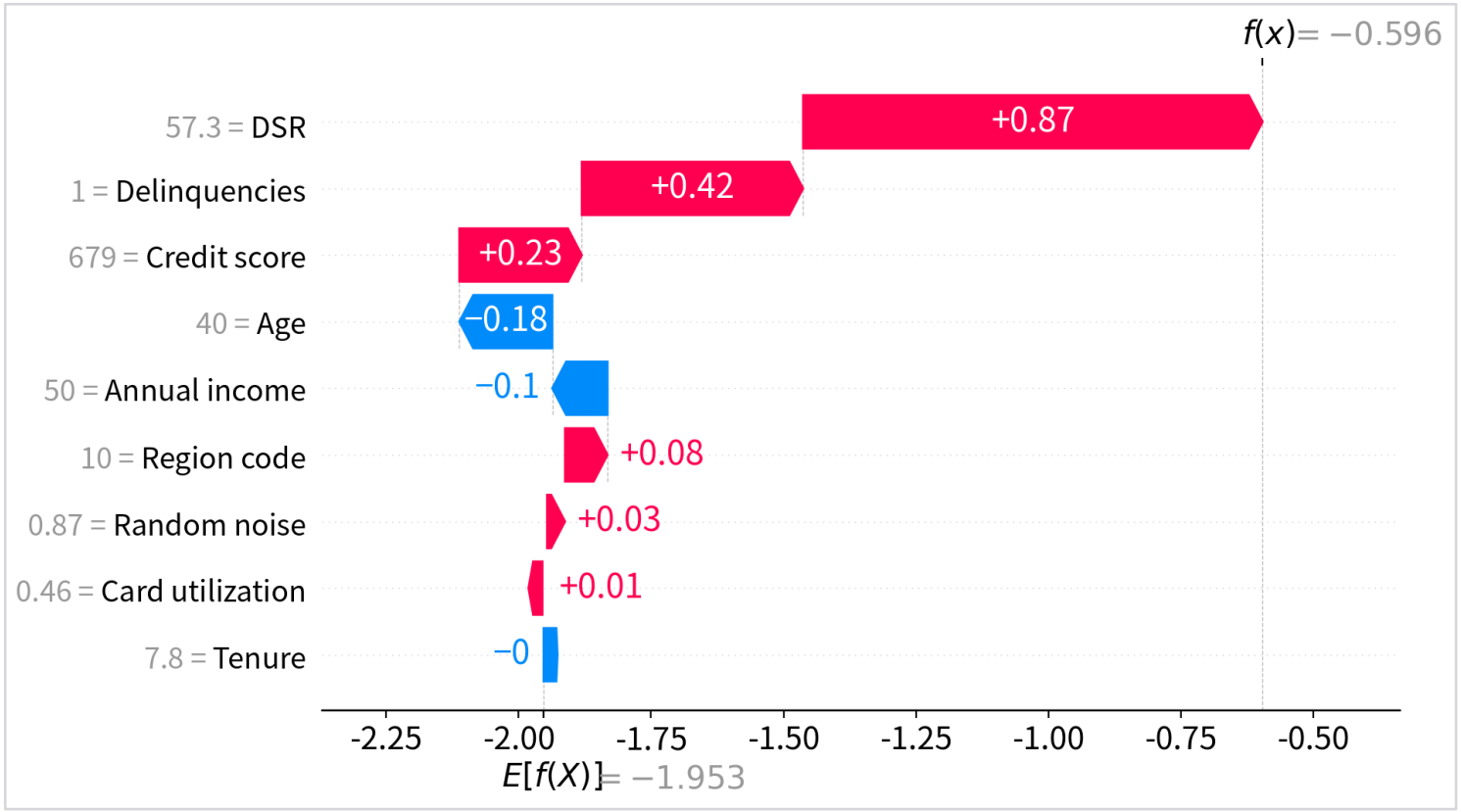
XGBoost and LightGBM have SHAP built in — predict(..., pred_contribs=True)

Decomposing One Customer's Default Prediction (Waterfall)

Synthetic credit model · one customer rejected at a 35.5% default probability — units are log-odds

How to read

- Bottom E[f(X)] = −1.953: the average customer (default probability 12.4%)
- Red factors raised risk, blue factors lowered it
- Sum f(x) = −0.596 → default probability 35.5%
- Region code +0.08: even unrelated variables pick up small contributions → maintain a list of variables excluded from reasons



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PART 5 · Lab

Reading a Model with XGBoost + SHAP

View global and individual effects together with summary (beeswarm) and dependence (scatter) plots

Code — From Model Training to Four Plots

Recommended API for shap 0.52 · XGBoost 3.x as of September 2026

[Code] Python

```
import shap, xgboost as xgb
model = xgb.XGBClassifier(n_estimators=400, max_depth=4).fit(X, y)
explainer = shap.TreeExplainer(model)       # exact computation for trees
sv = explainer(X)                           # Explanation object
shap.plots.beeswarm(sv)                     # summary plot (formerly summary_plot)
shap.plots.scatter(sv[:, "Age"], color=sv)  # dependence plot (formerly dependence_plot)
shap.plots.waterfall(sv[0])                 # decompose one case
shap.plots.bar(sv)                          # global importance = mean(|SHAP|)
```

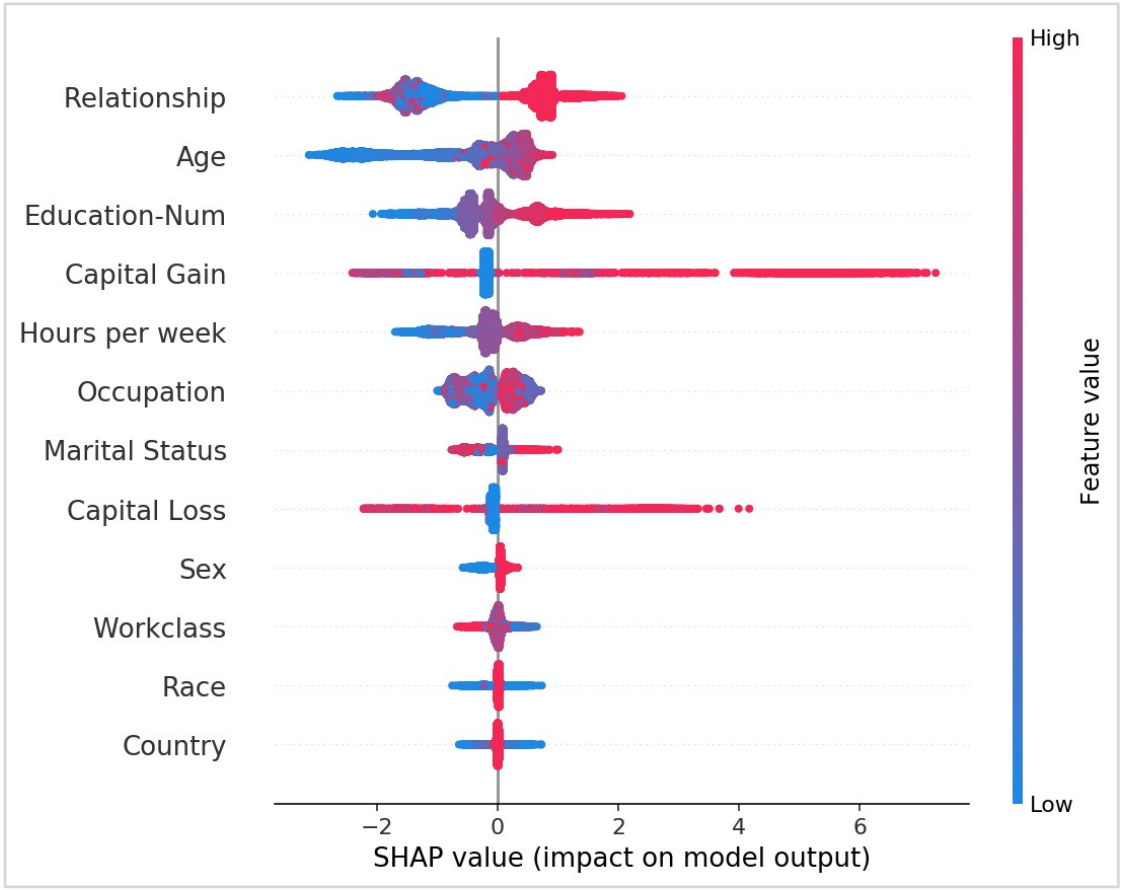
Original code shap.TreeExplainer(model).shap_values(X) → summary_plot(...) still works but is legacy. The new API draws every plot from a single Explanation object.

SHAP Values — Summary Plot

SHAP values are both consistent and additive — one plot holds the global ranking and individual distributions

How to read

- One dot = one person; horizontal position = that feature's SHAP value (log-odds)
- Color = feature value (red high · blue low)
- Vertical order = global ranking by mean(|SHAP|): Relationship → Age → Education-Num
- Capital Gain: mostly near 0, but a very large + effect for a few — a distribution invisible in bar importance



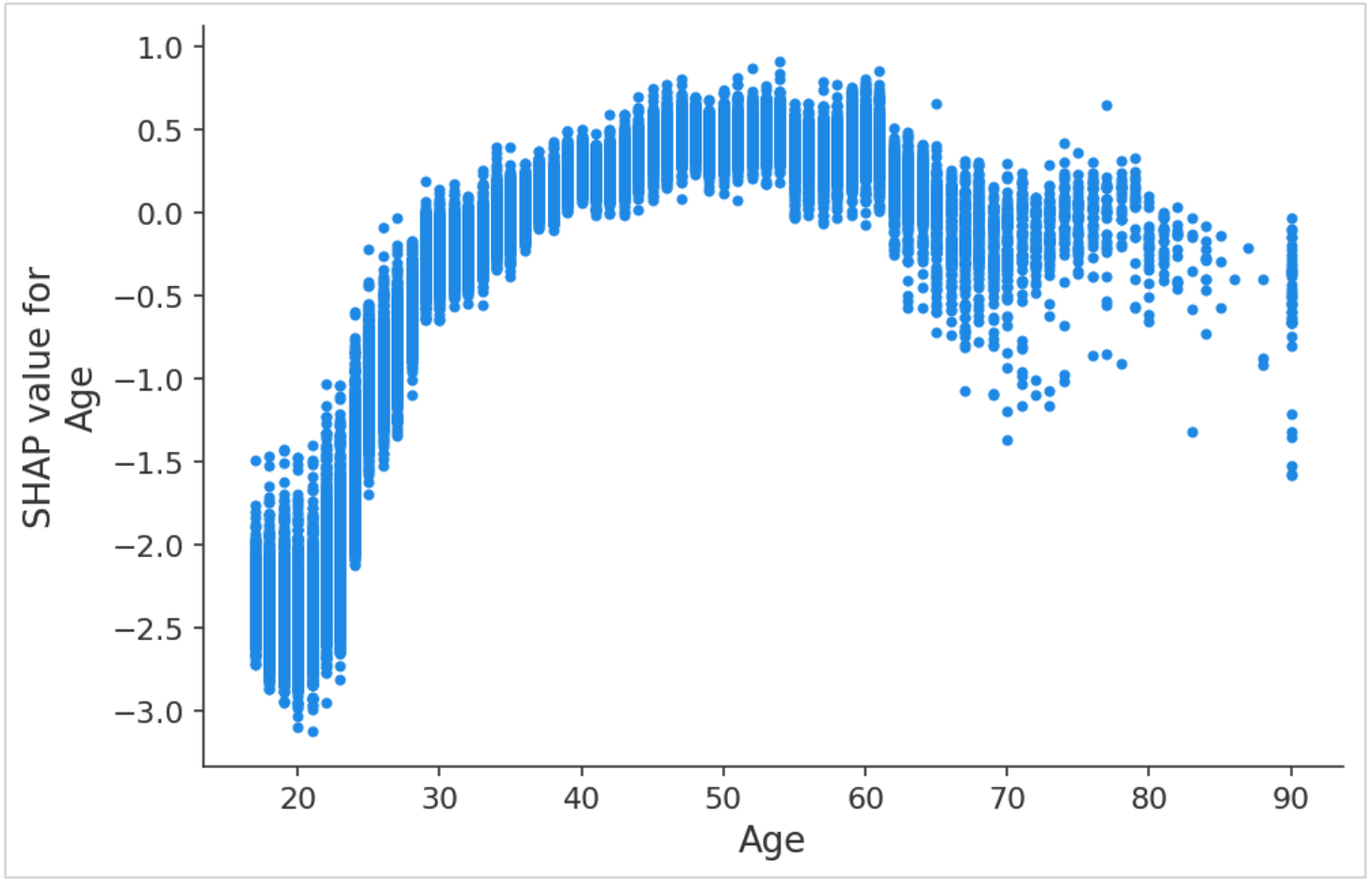
Original code: shap.summary_plot(shap_values, X)

Features and SHAP Values ① Age

Each dot is one person in the dataset. See how the income prediction changes with age

How to read

- y-axis: Age's contribution to the log-odds of "annual income above $50K" (more precise than the original "change in log odds ratio")
- Early 20s at −2 to −3: lowers the odds to about 1/10 (e^−2.3 ≈ 0.1)
- Peaks at ages 40–60, then declines gently
- Vertical spread at the same age = interaction with other features

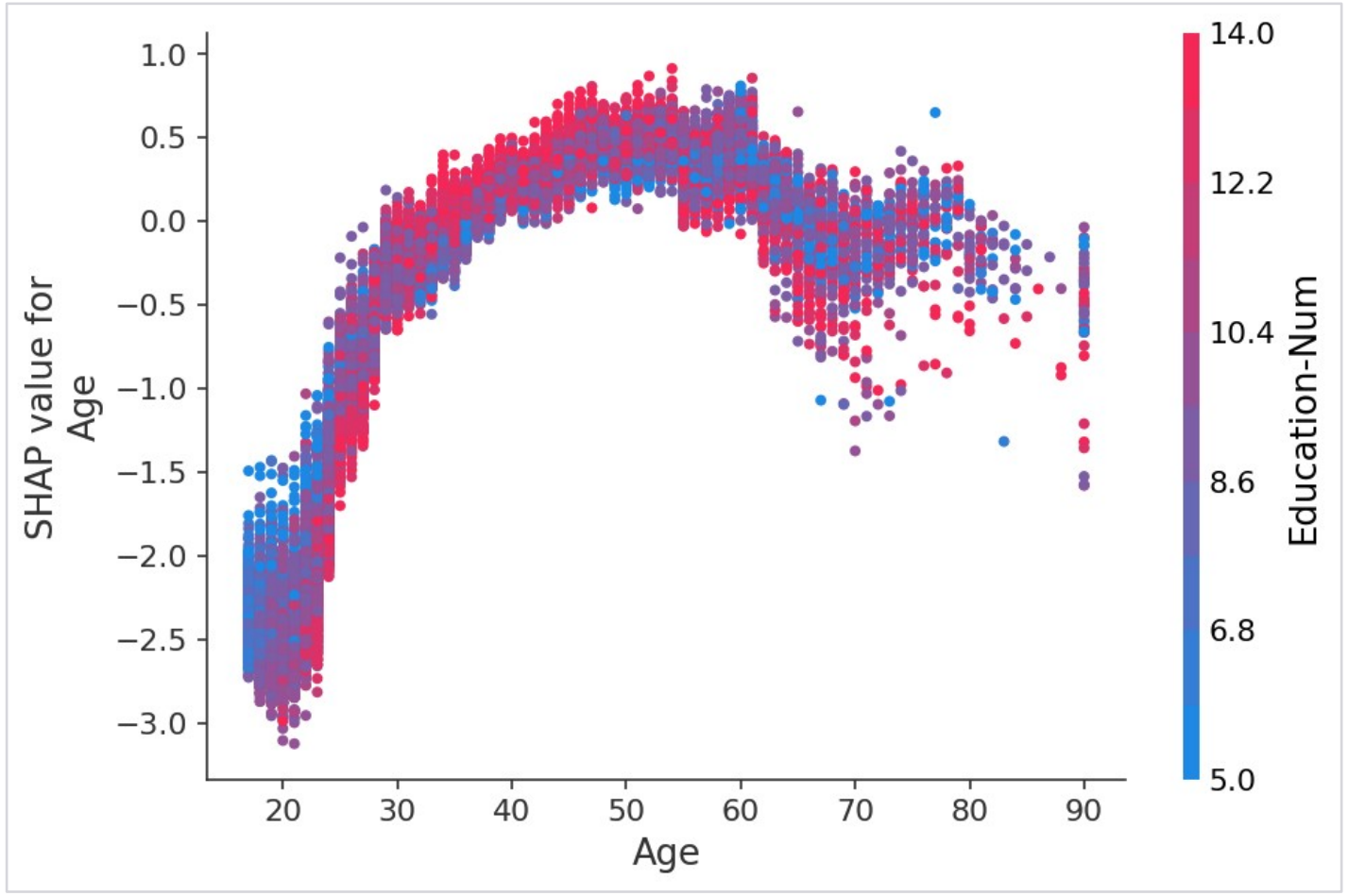


Features and SHAP Values ② Age × Years of Education

The same plot colored by years of education (Education-Num, years of schooling completed)

How to read

- Ages 30–50: red (more years of education) is generally higher — the age effect grows
- Around age 70, some red points drop lower
- If color explains the vertical spread, it signals an interaction between the two features

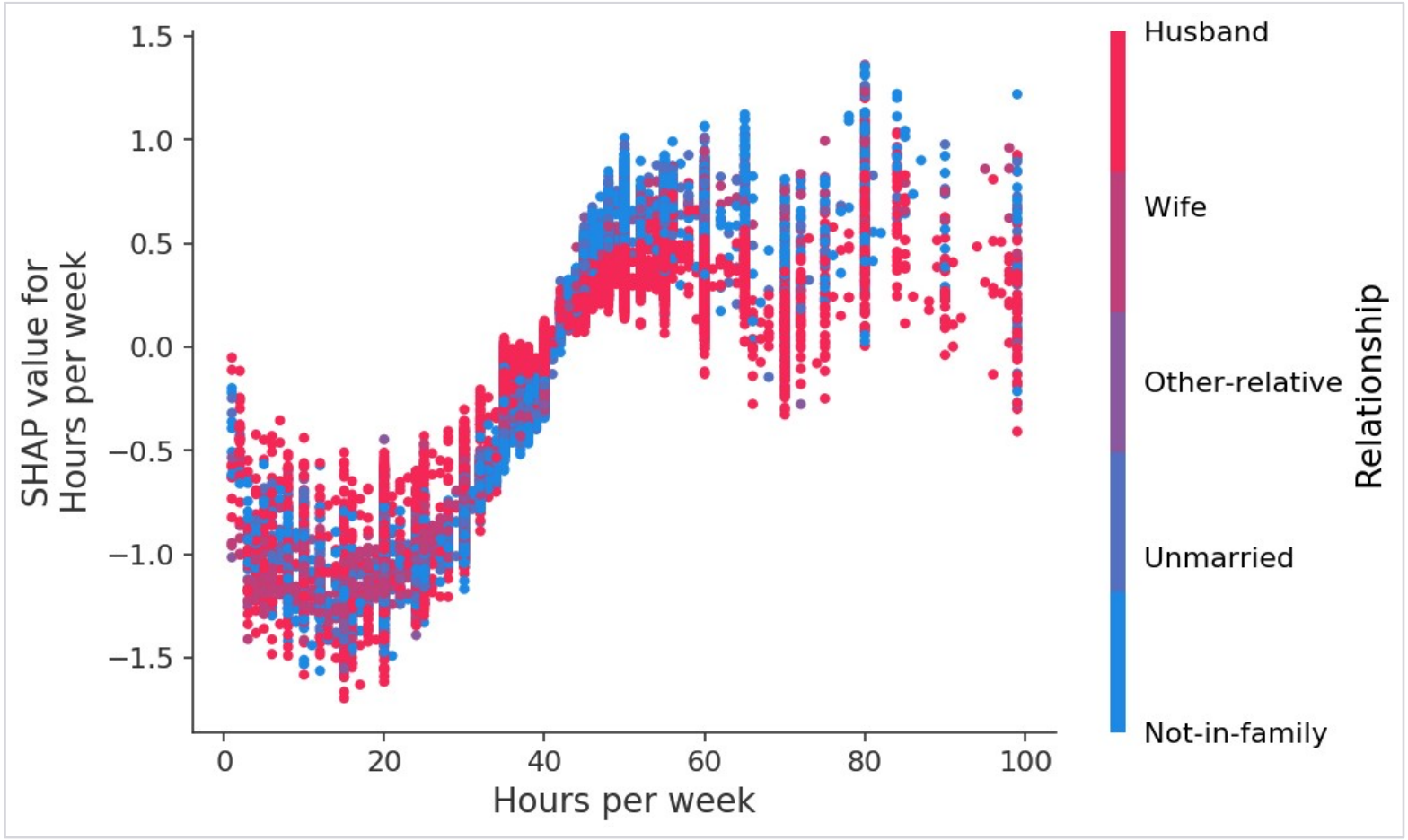


Features and SHAP Values ③ Hours per Week × Household Relationship

The effect of weekly working hours on the income prediction — it varies by household relationship (Relationship)

How to read

- Under 40 hours is −, turning + at 40–50 hours
- Beyond 50 hours the effect stops growing (saturation)
- 40–60 hours: blue (not-in-family · unmarried) is above red (husband) — the effect of extra work differs by household type

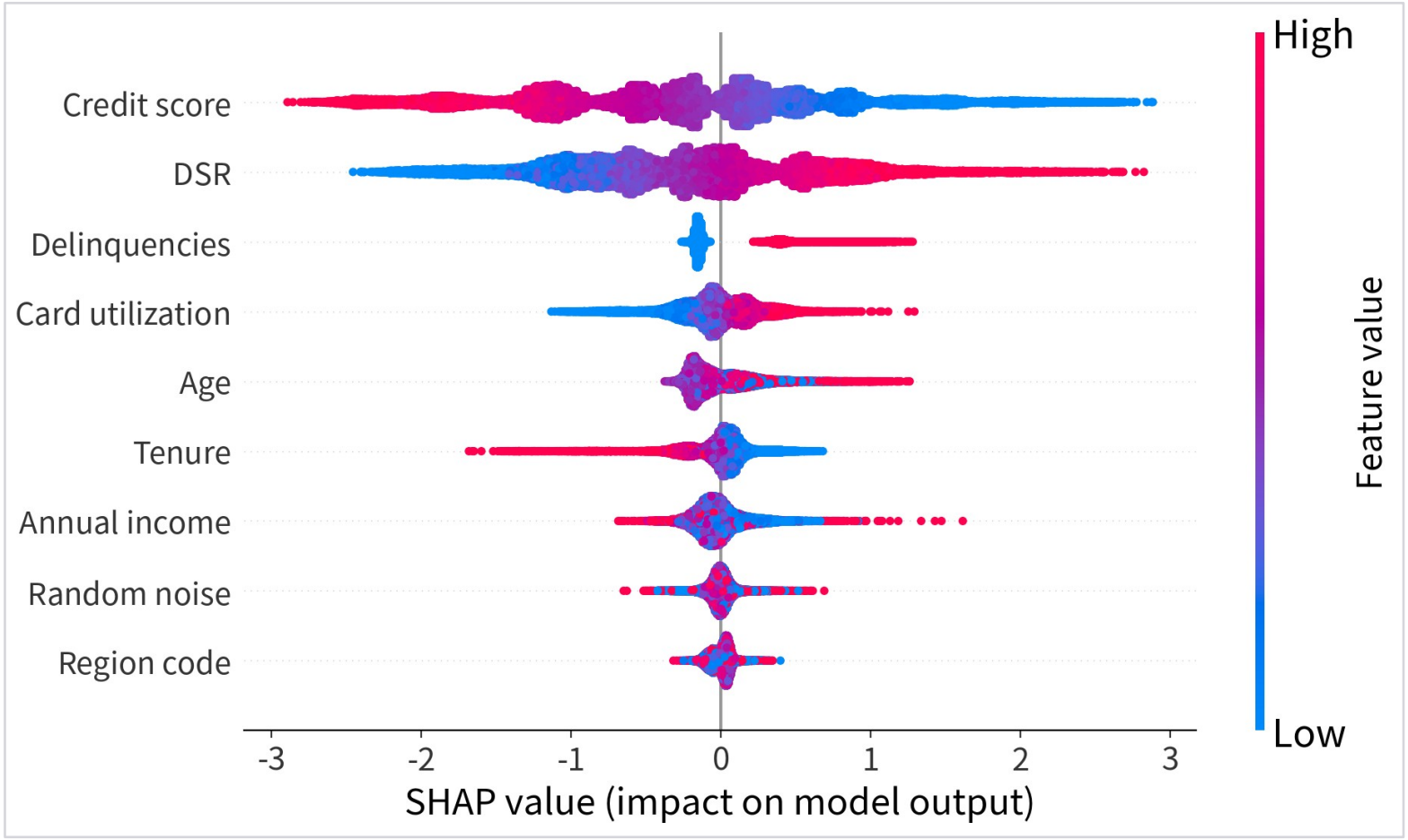


Summary Plot of the Synthetic Credit Data

Check whether SHAP recovers the direction of the true model on data with a known answer

What to check

- Credit score: lower (blue) → right = higher risk — same direction as the true model
- Delinquencies: slightly − at 0, clearly + at 1 or more (two clusters)
- Age: + at both ends → the true model's U-shaped effect
- Random noise · region code: clustered near 0

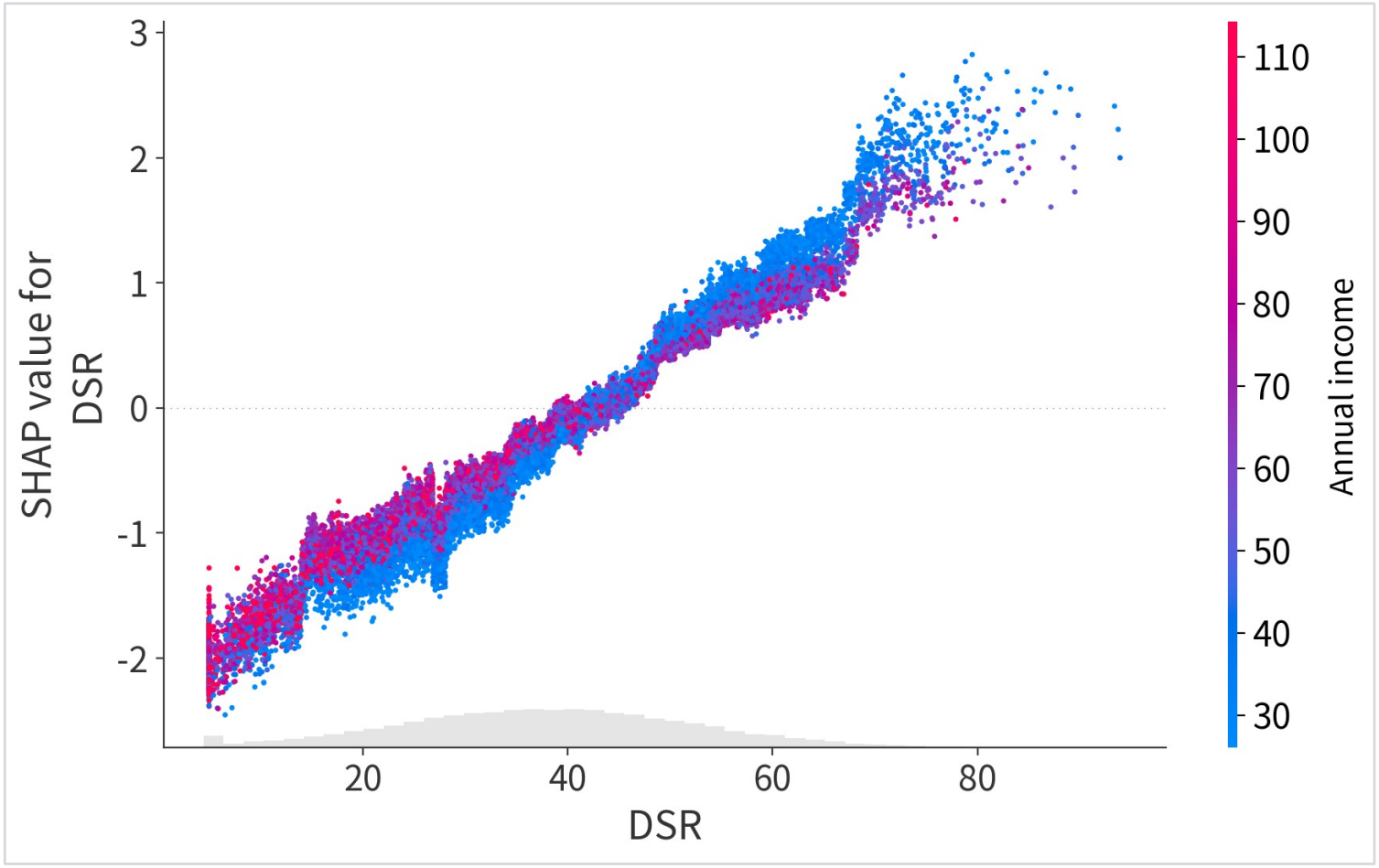


Finding a Hidden Interaction — DSR × Annual Income

The true model scaled the DSR effect by 1.8× for annual income below KRW 40 million. Will the SHAP dependence plot find it?

How to read

- DSR 40% and above: blue (low income) is higher — more risk at the same DSR
- DSR below 40%: blue is lower — the steeper slope fans out on both sides
- Vertical spread split by color = a hidden interaction



06

PART 6 · Advanced

Advanced SHAP — Interactions and Monitoring

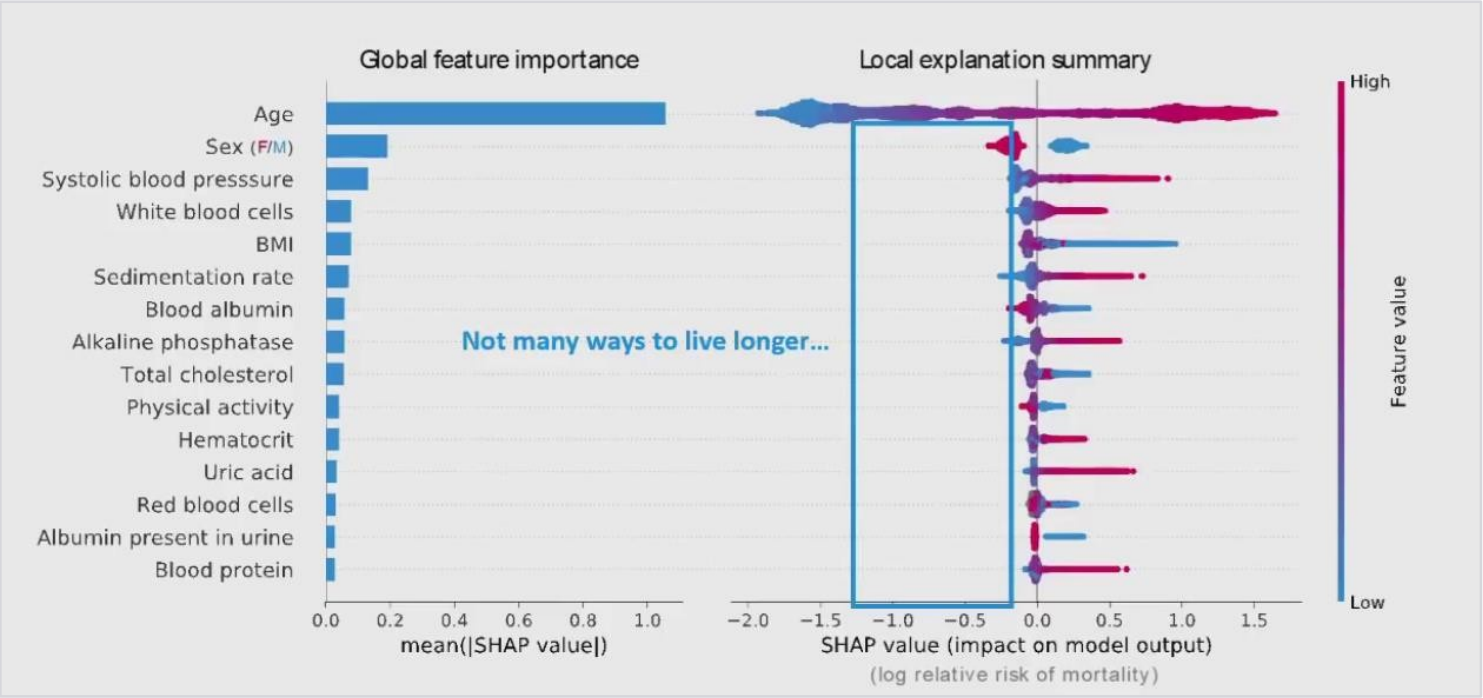
NHANES I mortality model and hospital surgery duration model (Lundberg et al., Nature Machine Intelligence 2020)

Discovering What Global Importance Misses

Computing feature contributions for each individual (local) reveals the distribution the average bar hides

How to read

- Data: NHANES I, about 14,000 U.S. adults · 20-year mortality follow-up
- Left bars = mean |SHAP| of the dots on the right
- "Not many ways to live longer": the left tail that lowers risk is short; the right tail that raises it is long
- Rare but large risks (e.g., high blood pressure) get buried in average importance

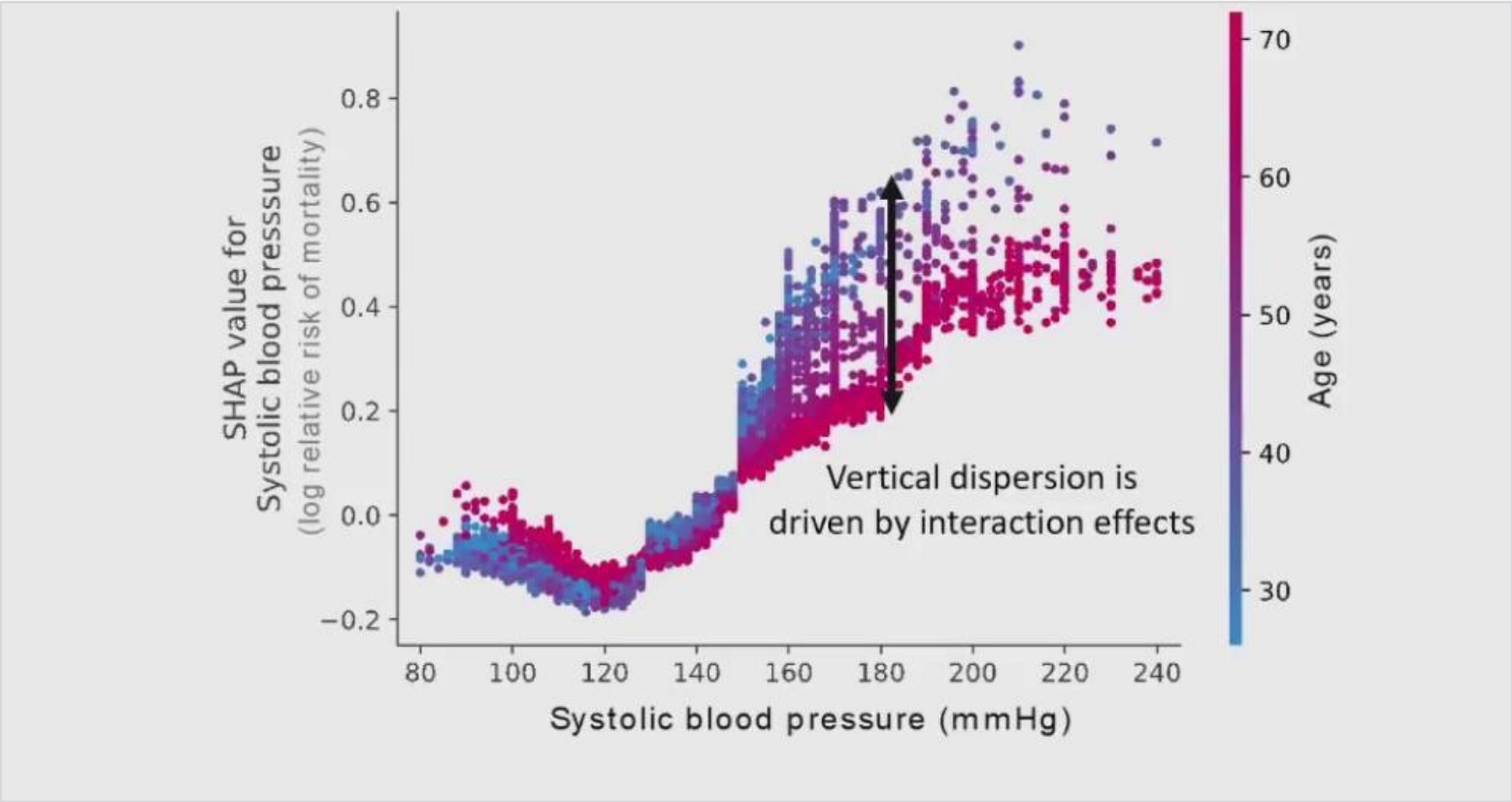


Two Features Can Be Considered Together

Dependence plot of systolic blood pressure SHAP values, colored by age

How to read

- Lowest near 120 mmHg; above that, risk rises with blood pressure
- Vertical spread at the same blood pressure = interaction with age
- 160–180 mmHg: blue (younger people) is higher — the model sees early-onset hypertension as riskier

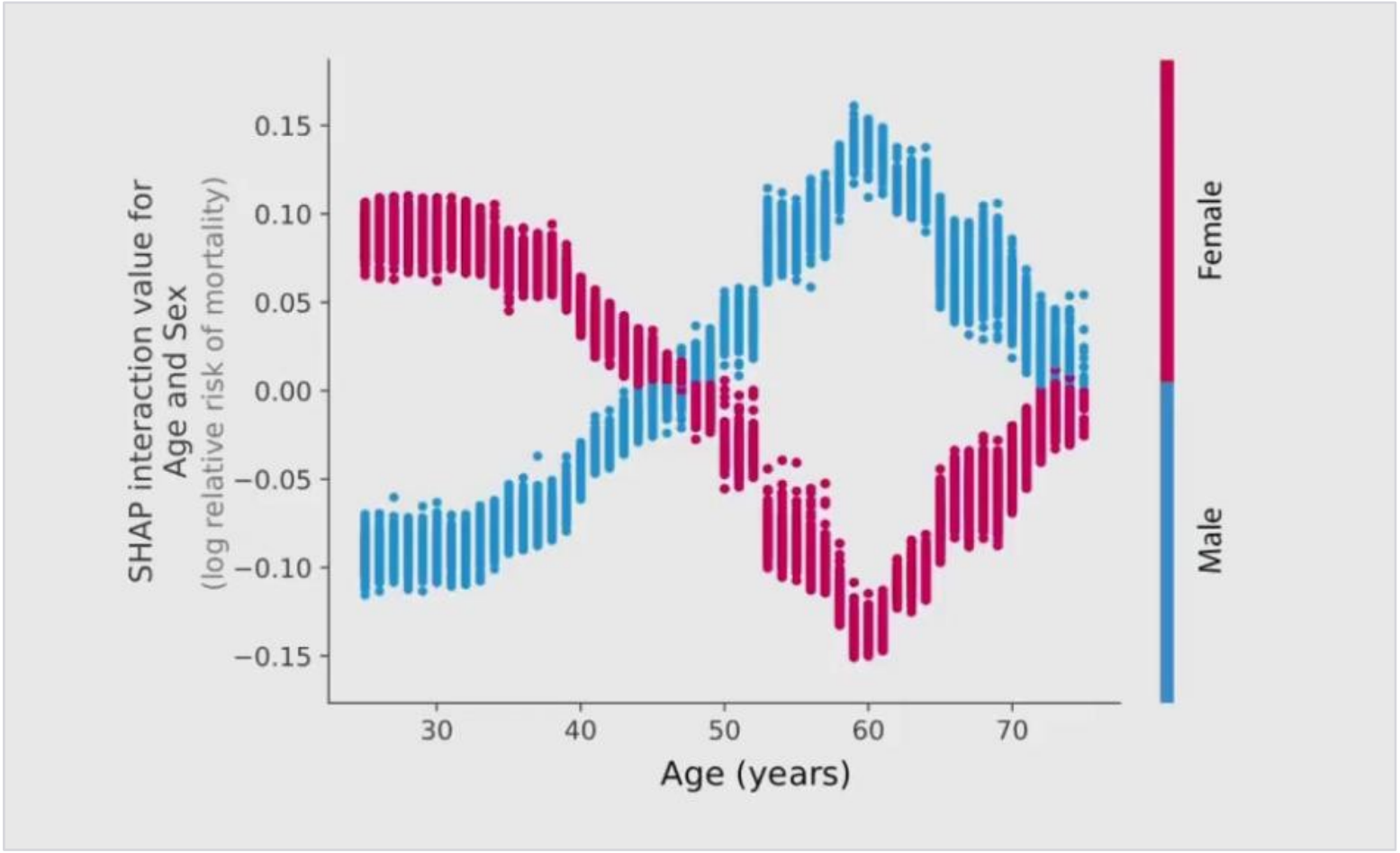


Decomposing One Feature and Considering It Jointly

SHAP interaction values: isolate only the part of the age effect entangled with sex

How to read

- When young: women +, men –
- Around age 60: men at maximum +, women at maximum – — the direction flips
- Removing this share leaves the pure main effect of age



Definition of SHAP Interaction Values

With the Shapley interaction index, contributions split exactly into main effects and pairwise interactions

$$\Phi_{i,j} = \sum_{S \subseteq N \setminus \{i,j\}} \frac{|S|!\,(M - |S| - 2)!}{2\,(M-1)!}\,\nabla_{ij}(S)$$

$$\nabla_{ij}(S) = v(S \cup \{i,j\}) - v(S \cup \{i\}) - v(S \cup \{j\}) + v(S), \qquad \phi_i = \Phi_{i,i} + \sum_{j \neq i} \Phi_{i,j}$$

How to read

- ∇: the extra amount that arises only when i and j are present together
- Φ(i,i) is the main effect; the rest are pairwise interactions

Code

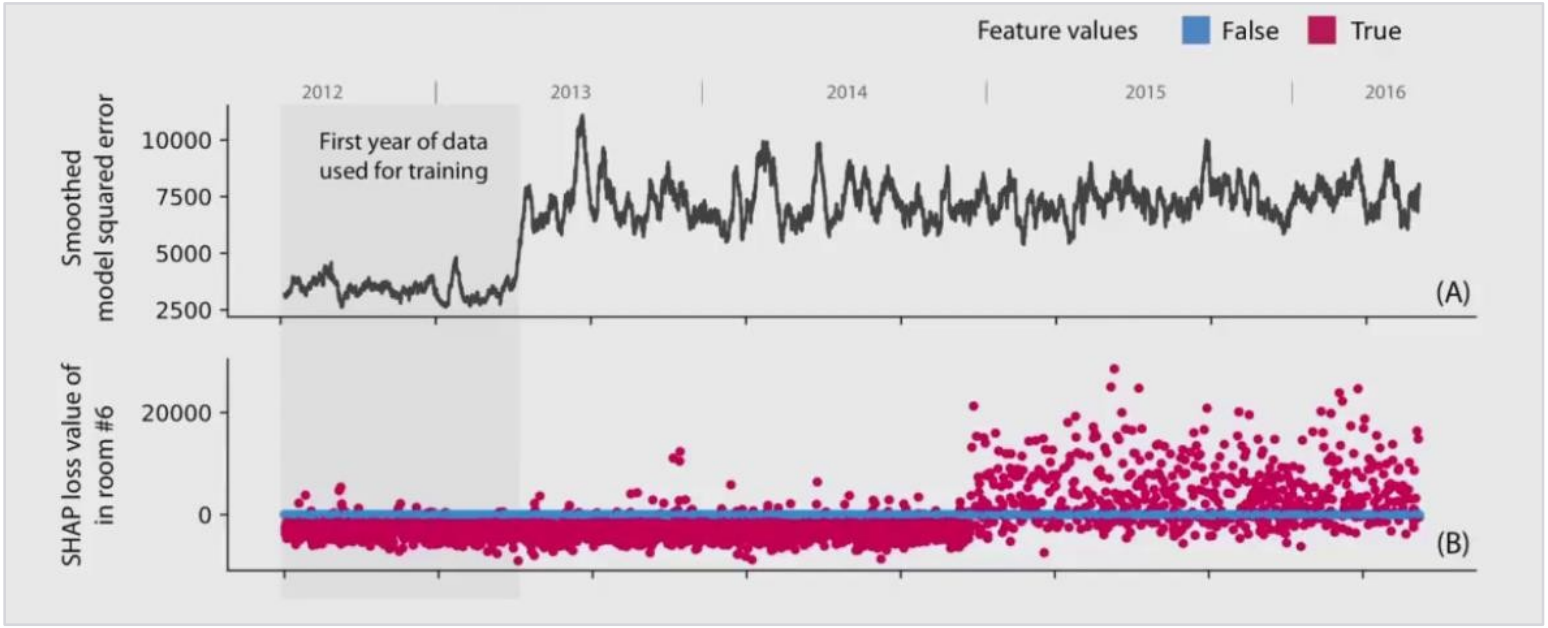
- explainer.shap_interaction_values(X)
- XGBoost: pred_interactions=True

Monitoring Models over Time ①

A surgery duration prediction model — track per-feature SHAP loss instead of overall error

What to look at

- (A) Overall loss: error grows after the training period, but the cause is unknown
- (B) The SHAP loss of the "operating room 6" feature grows from a certain point
- In fact, the researchers deliberately swapped the labels of rooms 6 and 13 as a test — finding an error invisible in overall loss

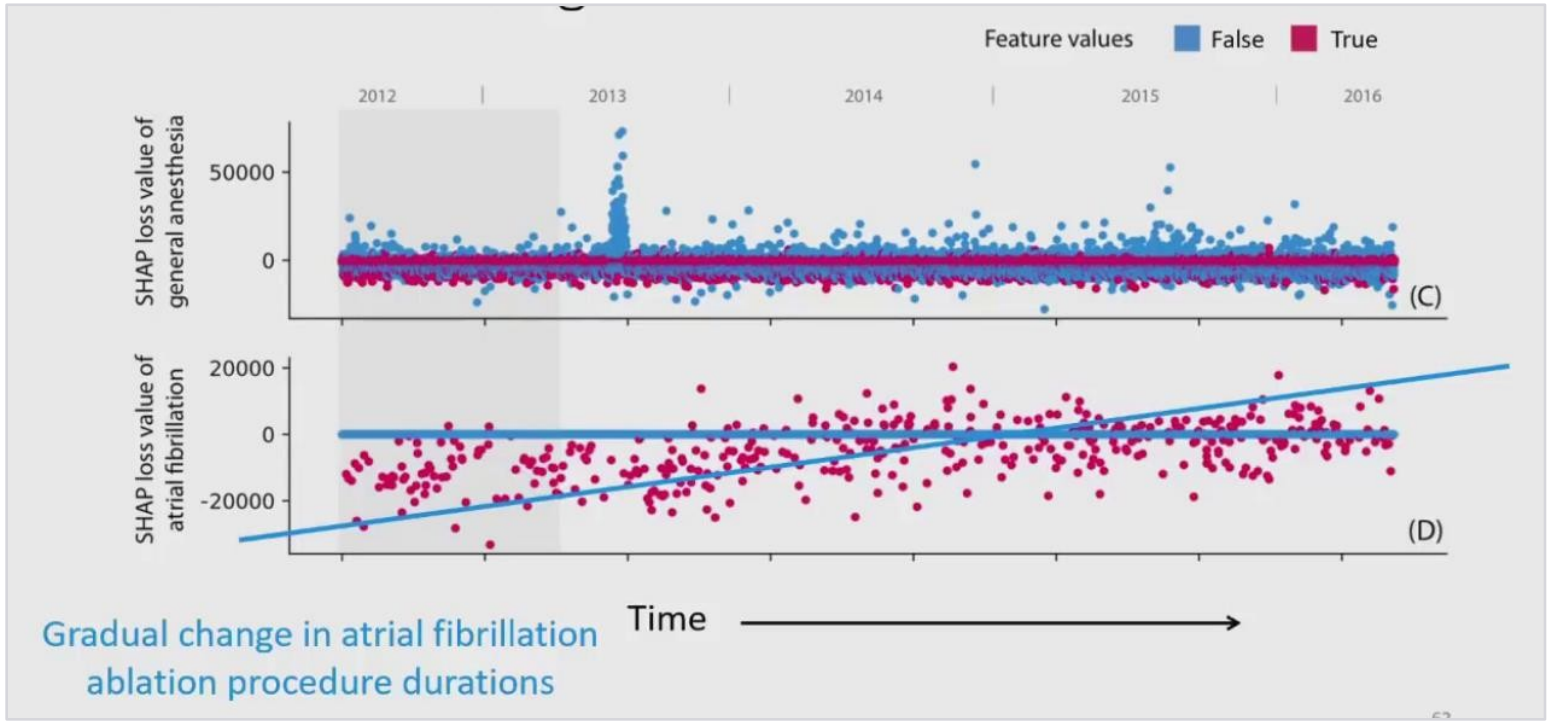


Monitoring Models over Time ②

Pinpoint the cause and timing of a problem at the feature level

What to look at

- (C) General anesthesia feature: spikes in early 2013 → uncovered a real problem, an electronic medical record configuration error
- (D) Atrial fibrillation feature: shifts gradually → drift as technology and staffing changed ablation procedure times
- Knowing the causal feature tells you whether to retrain or fix the data



Carrying this over to financial model monitoring

Input distribution metrics (e.g., PSI) only say "it changed"; SHAP loss tells "how much it hurt performance"

Hospital case	Financial model counterpart	Practical action
Swapped labels (rooms 6 · 13)	Branch or product code mapping error	Per-feature SHAP loss alerts
Temporary configuration error (anesthesia)	Delay or missing data from external credit bureau feeds	Source data checks · temporary rules
Gradual drift (atrial fibrillation)	Shift in DSR effect from interest rate or business cycle changes	Retraining · champion-challenger comparison

Monitoring = what changed (distribution) + what hurt performance (SHAP loss)

07

PART 7 · Financial Practice

Financial Applications and Limits of SHAP

Rejection reasons · regulation · pitfalls · recent trends (as of September 2026)

Building Rejection Reasons with SHAP

SHAP values → standard reason codes → customer notice

① Set the baseline

Background data

Decide and document what the base value will be

e.g., recent approvals

② Decompose each case

TreeSHAP

Compute SHAP for each rejection and keep only the factors that raised risk

e.g., DSR +0.87

③ Map to reason codes

Top 3–4

Convert to standard wording and exclude variables that cannot be used as reasons

e.g., "Excessive DSR"

④ Validate · record

Stability · history

Check that similar customers get consistent reasons, and archive them

e.g., reasons by month

SHAP is a basis for the decision, not causation — "do this and you'll be approved" calls for counterfactual analysis

Regulatory Landscape ① Korea (as of September 2026)

The duty to explain credit decisions is already in law, and the AI Basic Act broadens it

Basis	In force	Explanation requirements
Credit Information Act Art. 36-2	2020.8	Right to request explanation of whether automated assessment was used, its result, main criteria, and underlying information; right to object, correct, and request recalculation
AI Basic Act	2026.1.22	Loan underwriting is high-impact AI (Art. 2(4)) — must establish ways to explain the final result, main criteria, and an overview of training data (Art. 34)
AI Guidelines for the Financial Sector (revised)	2026.6	Seven principles: governance · legality · auxiliary role · reliability · financial stability · good faith · security; management by risk level

Financial Services Commission press releases · Korea Law Information Center

Regulatory Landscape ② EU · U.S. (as of September 2026)

The demand to "explain why" remains, while timelines and formats are changing

Region	Basis	Key point
EU	GDPR Art. 22 · Art. 15(1)(h) + CJEU rulings	SCHUFA (2023): a credit score is itself an automated decision / D&B (2025): the procedures and principles actually applied must be made understandable
EU	AI Act (Reg. 2024/1689)	Credit scoring = high-risk (Annex III), right to explanation of individual decisions (Art. 86) — high-risk obligations apply from Dec 2027 under the amendment (2026/1744)
U.S.	ECOA · Reg B	Notice of specific principal reasons upon rejection (CFPB Circular 2022-03 withdrawn May 2025; the legal duty remains)
U.S.	SR 26-2 (2026.4)	Model risk management guidance replacing SR 11-7 — a risk-based approach

Limits and Pitfalls of SHAP

BIS (2025): explanation techniques can be inaccurate and unstable — explanations cannot substitute for trust

Pitfall	What is the problem?	Response
Correlated features	Contribution is split arbitrarily among correlated variables (DSR · income)	Grouped-variable SHAP, state the value function choice
Not causation	"Lower your DSR and you'll be approved" holds only by the model's standard	Separate counterfactual explanation and causal analysis
Dependence on the reference distribution	Values change when the background data changes	Fix and document the background data
Manipulability	Explanations alone can be dressed up to look plausible (Slack et al., AIES 2020)	Independent validation, audit of the model itself
Instability	Similar customers may get different reasons	Reason stability testing

Case · FICO Explainable ML Challenge (2018)

Instead of explaining a black box, they replaced it with an interpretable model from the start

Accuracy gap between the interpretable model and the black box: under 1%

Timeline

- FICO, with Google, UC Berkeley and others, held a "black box + explanation" competition on HELOC (home equity line of credit) data
- Duke University's Rudin team submitted an interpretable model and visualization tool instead of a black box
- Not the official winner, but received the FICO Recognition Award

The issue

- Rudin (2019): for high-stakes decisions, don't explain black boxes — use interpretable models
- The accuracy–interpretability trade-off does not always hold → the "?" in Part 1
- Practice: set a baseline with a scorecard or EBM first, and use black box + SHAP only when that falls short

Rudin & Radin, Harvard Data Science Review (2019) · Rudin, Nature Machine Intelligence (2019)

2024–2026 Trends

The unit of explanation is widening (interactions · foundation models), and supervisors demand controls that presume its limits

Trend	Content	Implications for finance
Higher-order interactions	shapiq (NeurIPS 2024): computes Shapley interactions of order 2 and above	Validating combined effects such as DSR × income
Tabular foundation models	TabPFN v2 (Nature 2025) — supports SHAP integration	The same explanation framework for small-data credit models
Supervisors' view	BIS FSI (Sep 2025) "Managing explanations": acknowledges the limits of explanation techniques, discusses allowing high-performance models given complementary controls	Design governance as explanation + controls
LLM interpretation	Mechanistic interpretability — MIT Technology Review 10 Breakthrough Technologies 2026	Explaining generative AI assistants for advisory and underwriting

Discussion Questions

Q1	May we tell rejected customers the top SHAP factors as is? How should we present "contribution" that is not causation?	**Explanation · consumer protection**
Q2	What should be the basis for splitting contributions between correlated variables such as DSR and income?	**Correlation · base value**
Q3	Should existing models whose variables were selected by gain importance be revisited?	**Feature importance**
Q4	1% accuracy vs. interpretability — what are our organization's criteria?	**Model governance**

Summary — Three Takeaways for Today

1

Explainability is not optional

As with wolves and huskies, we can trust a model only by seeing its reasons — and for credit decisions, the law demands reasons

2

SHAP instead of XGBoost's built-in importance

The unique allocation that does not wobble with split order — combining consistency and local accuracy

3

SHAP is a tool, not a conclusion

Know its limits on correlation, base values, and causation, then use it for rejection reasons, monitoring, and regulatory compliance

Score → contribution → reason → validation: design the whole explanation process

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